AI CAREERS FOR WOMEN

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Tourism & Hospitality Booking Dashboard and Revenue Insights

Project Domain

Excel with Copilot

A Collaborative Initiative

Submitted in partial fulfillment of the requirements of the AIWC Project

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3. Problem Statement

- The tourism and hospitality industry generates a large amount of data, but it is often messy and hard to understand. This makes it difficult for businesses to analyze trends, understand customers, and make correct decisions.
- This problem is significant because poor data analysis can lead to incorrect pricing, inefficient resource allocation, and missed business opportunities. It is challenging due to issues like missing values, inconsistent formats, and lack of proper visualization tools.
- It mainly affects hotel managers, analysts, and businesses in the hospitality industry.
- Existing approaches lack proper data cleaning and effective visualization tools.
- The project focuses on using Microsoft Excel and Copilot to clean data and create an interactive dashboard, limited to analysis and visualization only.

4. Objectives

- To analyze tourism and hospitality booking data to identify key trends in bookings, revenue, and customer behavior.
- To clean and organize raw data for accurate and meaningful analysis.
- To create an interactive dashboard using Microsoft Excel for easy visualization of insights.
- To understand factors affecting revenue such as room types, seasons, and customer preferences.
- To support better decision-making in pricing, marketing, and resource management.
- To use Copilot Agent to enhance data analysis and generate insights efficiently.

5. Proposed Solution

- The solution is to clean, organize, and analyze tourism and hospitality data to make it easy to understand and use.
- This approach solves the problem by converting raw data into meaningful insights through analysis and visualization.
- The methodology includes data cleaning (removing duplicates, handling missing values), data analysis using pivot tables, and dashboard creation using charts and KPIs.
- The tools used are Microsoft Excel and Copilot Agent for data analysis, interpretation, and assistance.

- The steps followed include data collection, data cleaning, data transformation, analysis, and dashboard development.
- The key benefits include better decision-making, clear visualization, improved understanding of customer behavior, and efficient resource management.
- The innovation is the use of Copilot Agent with Excel, making the process faster, more interactive, and user-friendly compared to traditional methods.

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop or PC — Intel Core i3 or higher, minimum 4 GB RAM (8 GB recommended)
- Operating System: Windows 10 / 11 or macOS
- Microsoft Excel (Microsoft 365 / Excel 2019 or later)
- Power BI Desktop — latest version (free download from Microsoft) — where applicable
- Microsoft Copilot — via Microsoft 365 subscription — where applicable
- AI Chatbot (ChatGPT / Google Gemini / Microsoft Copilot Web) — for prompt-based projects
- Internet connection — required for Copilot and AI Chatbot access
- Web browser — Google Chrome or Microsoft Edge (latest version)

B. Dataset

- The dataset was obtained from Kaggle, containing tourism and hospitality booking and revenue data.The data included details such as bookings, customer information, payment types, and revenue.
- Data cleaning was performed by removing duplicates, missing values to ensure accuracy and date formats were standardized to maintain consistency across the dataset. Pivot tables were created to summarize and analyze key metrics.
- Copilot Agent was used on the cleaned dataset to assist in data analysis and interpretation.

C. Project Timeline

A simple table showing planned tasks and expected completion dates.

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	*Week 1*
2	Data cleaning, preprocessing, and validation / Prompt design	*Week 2*
3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	*Week 2 – 3*
4	Dashboard / output build, testing, and review	*Week 3*
5	Report writing, GitHub upload, and video recording and submission	*Week 4*

7. Expected Outcomes / Conclusion

This project successfully analyzed tourism and hospitality booking data by cleaning, organizing, and transforming raw data into meaningful insights. Using Microsoft Excel and Copilot Agent, an interactive dashboard was created to visualize key metrics such as booking trends, revenue patterns, and customer behavior.

The results highlighted important patterns that can help businesses improve decision-making, pricing strategies, and resource utilization. The objectives of the project were achieved by converting complex data into clear and understandable visual insights.

Overall, this project adds value to the hospitality domain by demonstrating how data analysis and AI assistance can enhance efficiency, accuracy, and business performance.

8. References

- Taiga Saito, Akihiko Takahashi, Noriaki Koide, & Yu Ichifuji (2019). *Application of Online Booking Data to Hotel Revenue Management*. Journal of Revenue and Pricing Management.
- Stanislav Ivanov & Vladimir Zhechev (2012). *Hotel Revenue Management – A Critical Literature Review*. Tourism & Management Studies.
- Kaggle (2024). *Tourism and Hospitality Booking Dataset*. Retrieved from https://www.kaggle.com
- Microsoft Excel Documentation. (2023). *Data Analysis and Visualization Tools*. Microsoft Corporation.